

## Practice Problems: Linear Algebra

ECON 441: Introduction to Mathematical Economics

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### Exercise 4.2

1. Given  $A = \begin{bmatrix} 7 & -1 \\ 6 & 9 \end{bmatrix}$ ,  $B = \begin{bmatrix} 0 & 4 \\ 3 & -2 \end{bmatrix}$ , and  $C = \begin{bmatrix} 8 & 3 \\ 6 & 1 \end{bmatrix}$ , find:

(a)  $A + B$

(c)  $3A$

(b)  $C - A$

(d)  $4B + 2C$

2. Given  $A = \begin{bmatrix} 2 & 8 \\ 3 & 0 \\ 5 & 1 \end{bmatrix}$ ,  $B = \begin{bmatrix} 2 & 0 \\ 3 & 8 \end{bmatrix}$ , and  $C = \begin{bmatrix} 7 & 2 \\ 6 & 3 \end{bmatrix}$ :

(a) Is  $AB$  defined? Calculate  $AB$ . Can you calculate  $BA$ ? Why?

(b) Is  $BC$  defined? Calculate  $BC$ . Is  $CB$  defined? If so, calculate  $CB$ . Is it true that  $BC = CB$ .

4. Find the product matrices in the following (in each case, append beneath every matrix a dimension indicator):

(a)  $\begin{bmatrix} 0 & 2 & 0 \\ 3 & 0 & 4 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 1 \\ 3 & 5 \end{bmatrix}$

(c)  $\begin{bmatrix} 3 & 5 & 0 \\ 4 & 2 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

(b)  $\begin{bmatrix} 6 & 5 & -1 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 5 & 2 \\ 0 & 1 \end{bmatrix}$

(d)  $\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 7 & 0 \\ 0 & 2 \\ 1 & 4 \end{bmatrix}$